1 Introduction

In this laboratory work, we will observe inverting and non-inverting operational amplifiers and feed the circuits with different inputs by function generator and observe the behaviour of the circuit with oscilloscope. This report will include our observations and insights from the experiments.

2 Theory

An operational amplifier is an integrated circuit composed of transistors, capacitors, and resistors. It has five main ports: two for power, two for signal input, and one for output. The op-amp amplifies the input signals from the V+ and V- inputs, with the amplification factor determined by the values of the other components in the circuit. If the signal is input through V-, the output signal will be 180 degrees out of phase. Conversely, if the signal is input through V+, the output signal will be in phase with the input signal.

From the inverting OpAmp we get these equations stated below, for the referance you can look at the Figure 1.

$$\frac{V_{in}}{R_1} * R_2 = V_1 - V_{out} \tag{1}$$

$$\frac{V_{out}}{V_{in}} = \frac{-R_2}{R_1} \tag{2}$$

From the non-inverting OpAmp we get these equations stated below, for the referance you can look at the Figure 6.

$$V_1 = V_2 \tag{3}$$

$$V_{out} = \frac{V_{in}}{R_2} * (R_1 + R_2) \tag{4}$$

$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2} \tag{5}$$

3 Experiments

In this laboratory session, we will hold two different experiments. First experiment will be about how to build and use circuit with inverting operational amplifier while feeding the circuit with different input signals while analyzing the response of the circuit and second experiment will be about how to build and use circuit with non-inverting operational amplifier while feeding the circuit with different input signals while analyzing the response of the circuit with oscilloscope.

3.1 Experiment 1: Inverting Amplifiers

First we will construct the circuit shown in the Figure 1 to observe behaviour of the inverting operational amplifier.

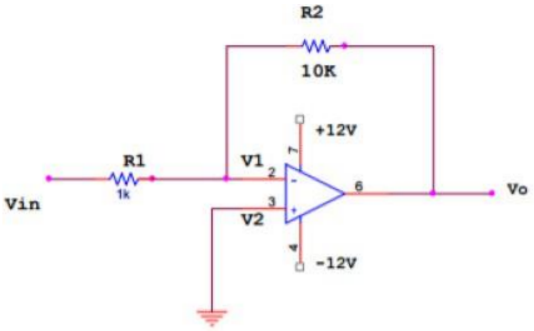
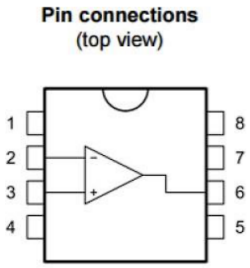


Figure 1: Inverting operational amplifier schematic.

In order to construct that circuit we need to use components listed below.

- 1X Breadboard
- 1X Function Generator
- 1X Oscilloscope
- 1X Power Supply
- 1X 741 IC OpAmp
- 1X 1kΩ
- 1X 10kΩ
- 5X Jumper cables

741 IC OpAmp schematic is shown in the Figure 2, to complete the circuit we are building we need to understand which pins of OpAmp is referring what.



1 - Offset null 1
2 - Inverting input
3 - Non-inverting input
4 - V_{CC}^-
5 - Offset null 2
6 - Output
7 - V_{CC}^+
8 - N.C.

Figure 2: Inverting operational amplifier schematic.

We have constructed the circuit shown in Figure 1 with inputs as +12V and -12V in series connection as shown in Figure 3.

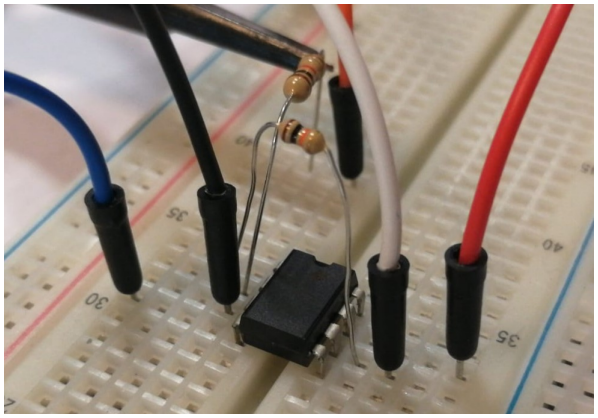


Figure 3: Inverting operational amplifier circuit on breadboard

As another input we have supplied with circuit with sinusoidal signal with 1kHz frequency and $1V_{pp}$ voltage from function generator and amplified that signal with amplifier circuit which we have built, then we had that output on oscilloscope as shown in the Figure 4.



Figure 4: First output from experiment 1

As we can see from the output our signals peak-to-peak voltage was given as $1V_{pp}$ and it is increased near to $10V_{pp}$ with our amplifier circuit, that is what we have been expecting from our circuit. Moreover their phases are not same, our output signal looks like it is shifted 180° to right.

After our measurements, then we have changed the settings of function generator, now we have supplied 1kHz frequency with increasing voltage to $2.5V_{pp}$.

After that adjustment our output changed unexpectedly as shown in the Figure 5.

From the previous measurements we are excepting to get our peak-to-peak voltage will be around $251V_{pp}$ but as we can see from the measurements and oscilloscope screen, our output signal is not perfect sinusoidal signal anymore and it is not providing our expectations in terms of peak-to-peak voltage, it is capped between some boundries.



Figure 5: Second output from experiment 1

These boundries are happening because our input voltage is +12V, with that given voltage inputs we can get up to 24 peak-to-peak voltage, since our input peak-to-peak voltage is $2.5V_{pp}$ our amplifier tries to exceed 24 volts but our circuit is not allowing to exceeding 24 volts.

3.2 Experiment 2: Non-inverting amplifiers.

In the previous experiment we have built inverting amplifiers, now we will build non-inverting amplifiers shown in the Figure 6 with same equipment.

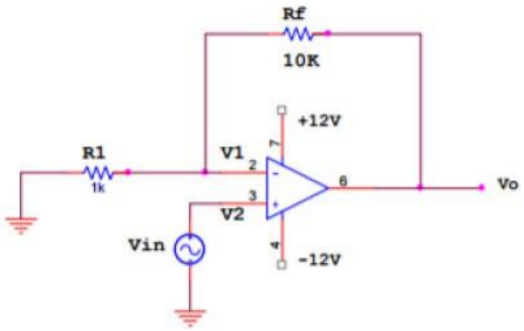


Figure 6: Schematic of the non-inverting amplifier

After we have built the circuit schematic shown in the Figure 6, we have supplied the circuit with same input as we did in the Experiment 1.

We have feed the circuit with +12V and -12V from power supply at first, then from the function generator we have supplied 1kHz frequency with $1V_{pp}$ voltage in sine form.

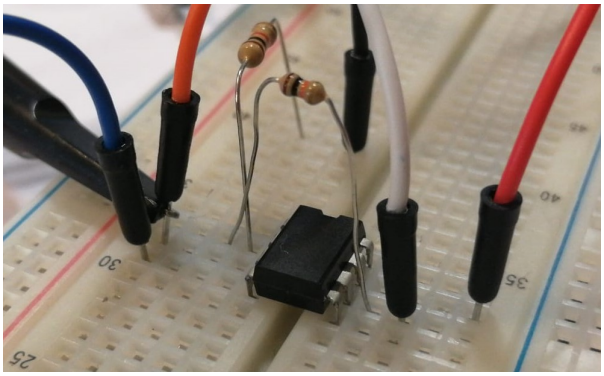


Figure 7: Circuit shown in the Figure 6 built on the breadboard

Our first outcome has shown in the Figure 8 below.



Figure 8: First output from experiment 2

As we can see from the output our signals peak-to-peak voltage was given as $1V_{pp}$ and it is increased near to $10V_{pp}$ with our amplifier circuit, that is what we have been expecting from our circuit. Moreover their phases are not same, our output signal looks like it is shifted 180° to right.

After our measurements, then we have changed the settings of function generator, now we have supplied 1kHz frequency with increasing voltage to $2.5V_{pp}$.

After that adjustment, our output changed unexpectedly as shown in the Figure 9.



Figure 9: Second output from experiment 2

From the previous measurements we are excepting to get our peak-to-peak voltage will be around $251V_{pp}$ but as we can see from the measurements and oscilloscope screen, our output signal is not perfect sinusoidal signal anymore and it is not providing our expectations in terms of peak-to-peak voltage, it is capped between some boundries.

These boundries are happening because our input voltage is +12V, with that given voltage inputs we can get up to 24 peak-to-peak voltage, since our input peak-to-peak voltage is $2.5V_{pp}$ our amplifier tries to exceed 24 volts but our circuit is not allowing to exceeding 24 volts.

4 Conclusion & Reflections

In our lab, we explored how operational amplifiers work by building and testing inverting and non-inverting circuits. We learned that our initial voltage supplied from power supply is critical for the boundaries of the amplification by changing the input signals and observing the outputs, we saw how these components are crucial for controlling signals in electronics.

5 Questions

1. To get a positive and a negative voltage -at the same time- from the DC power supply, which mode should be chosen (Independent / Serial / Parallel).

   In our experiment to get both a positive and a negative voltage simultaneously from a DC power supply, we should choose the Serial mode. This mode allows you to connect two outputs in series to create a split supply with a positive or negative.

2. Write the equation that calculate the voltage output value with respect to the resistors values in Op-Amp

   In the section 2 Theory we have calculated our Voltage outputs for the inverting amplifier at equation (2)

   $$\frac{V_{out}}{V_{in}} = \frac{-R_2}{R_1} \tag{2}$$

   and for the non-inverting amplifier at equation (5)

   $$\frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2} \tag{5}$$

3. Find the resistors values that should be used to get a value of 2 volt on the output of Op-Amp if the input voltage is 1 volt

   To achieve an output voltage V_{out} of 2 volts from an operational amplifier with an input voltage V_{in} of 1 volt, we need a gain 2. This can be achieved in a non-inverting op-amp configuration since the gain is positive. The gain formula for a non-inverting op-amp is:

$$1 + \frac{R_1}{R_2} \tag{6}$$

   and if we equal it to 2 since it is asked for we get the equation (7)

$$R_1 = R_2 \tag{7}$$

   This means that the values of the feedback resistor R_2 and the input resistor R_1 should be equal. We can choose any value for R_1 as long as R_2 is the same.